

```

In[ ]:= Clear["Global`*"]
ρ[x_] := Exp[-x^2];(*权函数*)
{xmin, xmax} = {0, Infinity};(*积分上下限*)
nmax = 100;(*最大阶*)
CoeInte1 = ParallelTable[Integrate[x^n*ρ[x], {x, xmin, xmax}], {n, 0, nmax*3}]
(* $\int_{xmin}^{xmax} x^n \rho(x) dx$ *)
CoeInte = CoeInte1;
CoeInte = N[CoeInte1, 10 000];(*精度控制*)

```

kernel 14 (Local)

**General:**  $0.00218333^{124}$  is too small to represent as a normalized machine number; precision may be lost. [i](#)

kernel 15 (Local)

**General:**  $0.0265134^{205}$  is too small to represent as a normalized machine number; precision may be lost. [i](#)

kernel 9 (Local)

**General:**  $0.016005^{289}$  is too small to represent as a normalized machine number; precision may be lost. [i](#)

kernel 6 (Local)

**General:**  $0.0400878^{279}$  is too small to represent as a normalized machine number; precision may be lost. [i](#)

kernel 1 (Local)

**General:**  $0.058991^{274}$  is too small to represent as a normalized machine number; precision may be lost. [i](#)

During evaluation of In[119]:=

**General:** Further output of `1` is too small to represent as a normalized machine number; precision may be lost. will be suppressed during this calculation. [i](#)

```

In[ ]:= (*
   $\psi_0(x)=1;$ 
   $\psi_k(x)=x^k+\sum_{j=0}^{k-1}c_{kj}\psi_j(x), k=1,2,\dots,n;$ 
   $c_{kj}=-\frac{(x^k,\psi_j)}{(\psi_j,\psi_j)}, (j=0,\dots,k-1)$ 
*)
nmax = 50;
psi = Table[0, {i, 0, nmax}, {j, 0, nmax}];(*基矢*)
nor = Table[0, {n, 0, nmax}];(*基矢内积、用作归一化*)
nor[[1]] = CoeInte[[1]];(*初始化*)
psi[[1, 1]] = 1;(*初始化*)
Clear[x]
For[k = 1, k ≤ nmax, k++,
  psi[[k + 1, k + 1]] = 1; (* $\psi_k(x)=x^k+\dots$ *)
  For[j = 0, j ≤ k - 1, j++,
    xkpsij = 0; (* $(x^k, \psi_j)$ *)
    For[i = 0, i < Length[psi[[j + 1]], i++,
      xkpsij = xkpsij + CoeInte[[i + k + 1]]*psi[[j + 1, i + 1]];
      psi[[k + 1]] = psi[[k + 1]] - psi[[j + 1]]*
         $\frac{xkpsij}{nor[[j + 1]]}$  (* $\psi_k(x)=x^k+\sum_{j=0}^{k-1}c_{kj}\psi_j(x), k=1,2,\dots,n;$ 
       $c_{kj}=-\frac{(x^k,\psi_j)}{(\psi_j,\psi_j)}=-\frac{xkpsij}{nor[[j+1]]}$  *)];
  ];
  For[i = 0, i < Length[psi[[k + 1]], i++,
    For[j = 0, j < Length[psi[[k + 1]], j++,
      nor[[k + 1]] = nor[[k + 1]] + CoeInte[[i + j + 1]]*psi[[k + 1, i + 1]]*psi[[k + 1, j + 1]];];
  ];
psi = Table[psi[[i]]/Sqrt[nor[[i]]], {i, 1, Length[psi]}];(*归一化*)
psi[i_, x_] := Expand[Sum[psi[[i + 1, j]]*x^(j - 1), {j, 1, Length[psi[[i + 1]]}]];(*产生正交多项式*)

```

```

In[ ]:= (*
  计算积分  $\int_{x_{min}}^{x_{max}} f(x)*\rho(x)dx$ ;
   $\sum_{k=0}^{n-1}A_k*x_k^m = \int_{x_{min}}^{x_{max}} x^m*\rho(x)dx$ ;  $m=0,1,\dots,n-1$ 。(n-1次代数精度)
   $x_k$ 是求积结点,  $A_k$ 是求积系数;
  则  $\sum_{k=0}^{n-1}A_k*f(x_k) = \int_{x_{min}}^{x_{max}} f(x)*\rho(x)dx$ 
*)
n = nmax - 0;
precision = 40;
Clear[x];
sol = Solve[psi[n, x] == 0, x];
xk = x /. sol;(*xk*)
xk = Re[N[xk, precision]];
b = Table[CoeInte[[m + 1]], {m, 0, n - 1}];(* $\int_{x_{min}}^{x_{max}} x^m*\rho(x)dx$ *)
M = Table[(xk)^m, {m, 0, n - 1}];(* $x_k^m$ *)
Ak = LinearSolve[M, b];

```

```

In[ ]:= Ak

```

```

Out[ ]:= {0.0094907437046997314708259619501929645244,
  0.022024804243228255363157406990994345041,

```

```

0.034374412277947122790940638861478112115,
0.046295442331579543397980063103425426984,
0.057427067352410260222871894066491281877,
0.067261674334723274831244343777561308734,
0.075165510542760654925043623502480092524,
0.080448721018930360298966675621565907590,
0.082486665378029384945925274396660416500,
0.080877284147973258830929950728585266969,
0.075596813121437394684006747674038304155,
0.067098896909613548203627627468923262987,
0.056303751410192878147438642131729632003,
0.044453364727085275320623179101316314462,
0.032860265552944118689549453475366179369,
0.022628104702244104191265108315299152919,
0.0144421205638529694046767627346418112257,
0.0084998891744169806356441465248401466048,
0.0045898158294769149510685134307628283039,
0.0022624803215574630035738851197747825643,
0.00101296345857313001359815884595248120994,
0.00040985505102577987580805844429250856481,
0.000149103396554347114376812913301943466035,
0.000048520972503506614540706161818063814958,
0.0000140499187954310437190543504800857238081,
3.6005744558412923094208715730869546977 × 10-6,
8.1205488566054141234696052443534759304 × 10-7,
1.60236226323029379082640999657464070570 × 10-7,
2.7491449821097336318562113004592676106 × 10-8,
4.0739387068860436224795148293412554695 × 10-9,
5.1773390940542655838147317132909720959 × 10-10,
5.5988313722907118332251748801829940407 × 10-11,
5.1083344028052837050676135164875501877 × 10-12,
3.8953118086079148380338775306535932781 × 10-13,
2.4563170831570278922204857987117383733 × 10-14,
1.26561330931793728121422349464643717198 × 10-15,
5.255826533367613737189187295118160852 × 10-17,
1.73149373672351541870678717161253105767 × 10-18,
4.4419824036643493904550771389771872023 × 10-20,
8.6801390035967722673358110118585334334 × 10-22,
1.25805027635560521880546114372124277045 × 10-23,
1.30872128650189205690216944494775473784 × 10-25,
9.3765255361412026543179338925314890393 × 10-28,
4.3860415462163842480085207583969339763 × 10-30,
1.24690574347624824784185237216936366505 × 10-32,
1.94846954012880587823603078981443677134 × 10-35,
1.44054648468423228015782066546864111386 × 10-38,
3.9404980221823086953397171527396556140 × 10-42,
2.7160779951078289527476416180363325898 × 10-46,
-4.0694037045787301994120242939956556684 × 10-50}

```

In[ ]:= **xk**

```
Out[ ]:= {0.003699416689411893870783591680559113232908,
0.01946557855930012623773073854964373235299,
0.04772325751107755698844474997431423006217,
0.08829947414233261368105151274070524571562,
0.1409442498315128378310201148278609109405,
0.2053438247329751545365997540525061511707,
0.2811309447311559387912428954647005984162,
0.3678957018671008160920077315988415592775,
0.4651966323825674804668429999420565211313,
0.5725715758397391932663470054247059294453,
0.6895478642934006363146541068341444286900,
0.8156515241290354568703957656097213628819,
0.9504152937670518424874499765880996170085,
1.093385370852204961506286769371877863143,
1.244126893940477146637742249017857971477,
1.402228232528363534349810142985390145464,
1.567304205795908445862260508925242035406,
1.738998377329241384240148985720837177816,
1.916984584253675956306353178956063896574,
2.100967858866580985969611716300591459860,
2.290684892955381631233279510404276446598,
2.485904182872338862795302984051769197683,
2.686425979769292270087937575812273204364,
2.892082156166672584109653702676916232450,
3.102736088691436076921775156612466156791,
3.318282648426356722872514877036456391187,
3.538648385701126281393640142782579224380,
3.763791996101299295259210632862261750589,
3.993705159874530912613635915962306540835,
4.228413859001339389021176878741016108386,
4.467980296781769765272930933539135620572,
4.712505576633350364903162868007964095012,
4.962133344173209139612605710548402570780,
5.217054666250212948445444938438896383653,
5.477514522993292932755037065728330828075,
5.743820441241526011112931323598643231194,
6.016354028121806048689106489706917557261,
6.295586519829836834700357000232729999920,
6.582100026386510109275009200471137111158,
6.876617079656574881995394529560166120026,
7.180042665166922917956590384471640813829,
7.493525705142409634065999019422811377926,
7.818552150395740678893533741594589943100,
8.157092104501688780897447536785491161023,
8.511845264743718491401746088993372098417,
8.886680059244128978941543062788124378804,
9.287496741416486046542554972296669095676,
```

```

9.724165865884631460833296700570326678019,
10.21588625857842815222310203336323771244,
10.81298607294536085737197741648321154445}

```

```

In[ ]:= Clear[f];
f[x_] := Sin[-x];
precision = 40;
int1 = Sum[Ak[[i]] * f[xk[[i]]], {i, 1, Length[xk]}];
int2 = N[Integrate[f[x] * ρ[x], {x, xmin, xmax}], precision];
int3 = NIntegrate[f[x] * ρ[x], {x, xmin, xmax}];
int1
int2
int3
int1 - int3
int2 - int3

```

```
Out[ ]:= -0.42443638350202229593404235248966957110
```

```
Out[ ]:= -0.4244363835020222959340423524896695710964
```

```
Out[ ]:= -0.424436
```

```
Out[ ]:= 3.12972 × 10-13
```

```
Out[ ]:= 3.12972 × 10-13
```